

This homework is due on **Tuesday, Nov. 5, 2024 by 2:00pm**. Complete this assignment in your own notebook and then upload it as a single, fully executed PDF to Brightspace > Gradescope > Homework 3. Email submissions and/or submissions in any format other than PDF **will not be accepted**. You must clearly identify which question you are answering throughout your homework. If we cannot tell what question you are answering, we cannot give you credit. This homework is worth 55 points. Incorrectly formatted assignments will be considered late until the correct version is uploaded.

1. The file `shared/data/temperature.csv` contains the average monthly temperatures for various locations around the world, identified by their latitude and longitude. You can find the file in our shared folder. Please make sure you understand the definitions of latitude and longitude before starting the assignment.
 - (a) **(2 points)** Create scatterplots showing the relationship between temperature and latitude for January and July separately. Plot them on two different graphs. Calculate the correlation between temperature and latitude for January and July separately.
 - (b) **(1 point)** Is there a natural justification to choose one variable for the x-axis and the other for the y-axis? Explain your reasoning.
 - (c) **(1 point)** Does the correlation change depending on which variable is treated as X ? Explain why or why not.
 - (d) **(6 points)**. Describe the patterns you observe both within each scatterplot and between the two correlations (for January and July). What could explain these patterns? In your analysis, discuss the magnitude of the correlations.
 - (e) **(4 points)** Create a scatterplot of latitude (y-axis) and longitude (x-axis) of the 2000 weather stations. Based on the plot, does the distribution of weather stations appear to be a random sample of Earth's surface? Calculate the minimum and maximum latitudes in the dataset. How might the latitude range affect the conclusions about temperature correlations from earlier?
 - (f) **(2 points)** Make a scatterplot and compute the correlation between latitude and temperature in July for the locations in the southern hemisphere (latitude < 0) only. Repeat the scatterplot and correlation calculation for locations only in the Northern Hemisphere (latitude > 0) for January.
 - (g) **(4 points)** Describe your findings in the previous part. Describe the things you notice? Why do you think that is?
 - (h) **(2 points)** Choose one of the scatterplots you created earlier and improve its visual display in at least two ways. Explain your choices.

2. The file `shared/data/momiq.csv` contains a dataset with child test scores and the IQ score of their mothers. Please read the dataset as a `DataFrame` and answer the following questions.
- (a) **(1 point)** What is the formula for the slope of the regression line when predicting Y from X , in terms of the standard deviations of X (σ_x) and Y (σ_y), and their correlation ρ ?
 - (b) **(2 points)** Using the columns `mom_iq` and `kid_score`, select one as the explanatory variable X and the other as the response variable Y . Create a scatterplot. Why did you assign the variables X and Y in this way? What is the correlation between the two variables?
 - (c) **(2 points)** Write a function `standardize` that takes as input a column and returns the corresponding standardized values. Provide comments in your code as needed.
 - (d) **(3 points)** Using your `standardize` function, standardize both `mom_iq` and `kid_score`. Assign the results to two new columns. Recreate the earlier scatterplot, but now using their standardized versions instead. Set the x - and y -axis limits to the interval $[-3, 3]$. Comment on any similarities or differences between the original and standardized scatterplots.
 - (e) **(3 points)** Using `statsmodels.formula.api.ols`, fit a linear regression model that predicts the standardized Y from the standardized X . Provide the model summary. What do you notice about the slope of the regression line for the standardized data?
 - (f) **(3 points)** Recreate the same scatterplot from question (d), but this time add the following two lines that span the x -axis interval $[-3, 3]$.
 - The line $Y = X$, in **black**
 - The regression line, in **red**
 - (g) **(2 points)** Which line fits the data better? Provide a brief explanation of what you notice.

3. The exercise is inspired by the “Bread and Peace” model ([Hibbs, 2000](#)), an election model that predicts the proportion of the two-party popular vote (`PopVoteProp`) won by the incumbent party. We will start by looking at a predictor called `avg_recent_growth`, which measures average recent growth in personal income.
- (a) **(1 point)** Read the dataset `shared/data/bread_votes.csv` into a `DataFrame`. Report the mean and standard deviation of the column `avg_recent_growth`, rounded to two decimal places. Also, report how many observations (rows) the dataset contains.
- (b) **(2 points)** Commenting your code to indicate your thought process, create a visualization of the data in the following way:
- Create a simple scatterplot of the `avg_recent_growth` and `incumbent_party_share` columns.
 - Stepping through each row of the dataset using a `for` loop, use `plt.text` to display the corresponding year at the `(avg_recent_growth, incumbent_party_share)` coordinate.
 - Optionally, you can color the scatter points white so that only the year values are shown on the plot.
- (c) **(3 points)** Compute the correlation between `avg_recent_growth` and `incumbent_party_share` to three decimal places. Answer the following:
- What does the direction and strength of the correlation tell us about the relationship between these two variables, at least in the sample?
 - After a term with a significantly higher than average growth in personal income, which party tends to see an increase in vote share? Please provide a hypothesis for why this might occur.
- (d) **(4 points)** Using `statsmodels.formula.api.ols`, fit a simple linear regression model to estimate the incumbent party’s vote share based on the average recent income growth. Comment your code appropriately and explain what the slope and intercept represent in the context of this dataset.

- (e) **(3 points)** Suppose the `avg_recent_growth` for the 2024 election is 2.47%. Using your model, predict the incumbent party's popular vote share in 2024.
- (f) **(4 points)** The formula for the standard error of a prediction is given by:

$$SE_{\text{pred}} = s \sqrt{1 + \frac{1}{n} + \frac{(x_{\text{new}} - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

where:

- $s = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}}$ is the root mean square error (RMSE),
- x_{new} is `avg_recent_growth` for the new observation (2.47),
- $\bar{x}$ is the mean of `avg_recent_growth`,
- n is the number of observations in the sample, and
- $\sum_{i=1}^n (x_i - \bar{x})^2$ is the sum of squared deviations of `avg_recent_growth`

You can compute s using the model's residuals with `model.resid`. After using the formula above to compute the standard error of the prediction for x_{new} , please calculate a 95% confidence interval for the predicted $\hat{y}_{\text{new}}$ as:

$$[\hat{y}_{\text{new}} - 2 \cdot SE_{\text{pred}}, \hat{y}_{\text{new}} + 2 \cdot SE_{\text{pred}}]$$

and report the lower and upper bounds rounded to two decimal places. Comment your code appropriately along the way.

References

Douglas A. Hibbs. Bread and peace voting in u.s. presidential elections. *Public Choice*, 104:149–180, 2000. doi: 10.1023/A:1005292312412. URL <https://doi.org/10.1023/A:1005292312412>.